

Decoding Multimodal Deception: A Deep Learning Approach to Information Disorder

1. Introduction

1.1 Problem Statement

In the social media era, platforms like Reddit have transformed information consumption. Unlike traditional text-based falsehoods, modern misinformation increasingly utilizes multimodal forms — combining real images with deceptive text to leverage visual "realness". This phenomenon, termed Information Disorder (Wardle & Derakhshan, 2017), blurs lines between mis-, dis-, and mal-information. Based on the Visual Turn (Mitchell, 1994), this study analyzes how visuals drive emotional persuasion in post-truth communication. We implement a BERT+ResNet model using the Fakeddit dataset to interpret AI behavior through a communication lens.

1.2 Theoretical Frameworks

This study is grounded in two complementary theoretical frameworks drawn from communication studies.

First, the Visual Turn (Mitchell, 1994) contends that contemporary media culture has shifted from text-dominated discourse toward image-centric communication. In digital environments, visual elements carry epistemic and affective authority that frequently exceeds that of accompanying text. This asymmetry is deliberately exploited in multimodal misinformation: when a fabricated narrative is paired with an authentic photograph, the perceived "realness" of the image lends unwarranted credibility to the false claim — a strategy our model is designed to detect.

Second, the Post-Truth framework (Wardle & Derakhshan, 2017) describes an information environment in which audience beliefs are shaped more by emotional resonance and community identity than by factual verification. Platform algorithms on sites like Reddit amplify this dynamic by ranking content according to engagement metrics (score, upvote ratio) rather than factual accuracy, creating structural incentives for emotionally compelling but potentially deceptive multimodal content. Together, these frameworks motivate a multimodal detection approach: understanding deception requires attending to both the semantic content of text and the affective power of images.

1.3 Research Questions (RQ)

- **RQ1:** To what extent do visual cues enhance the deceptiveness of misinformation?
 - **RQ2:** How do patterns of multimodal misinformation vary across different Subreddits?
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2. Methodology

2.1 Problem Formalization & Data

- **Dataset:** Fakeddit is an appropriate dataset for this study because it provides fine-grained multimodal image-text pairs together with Reddit-native social metadata, including subreddit, score, and upvote ratio. This enables misinformation to be analyzed not only as misleading content, but also as a platform-specific social phenomenon, As shown in Figure 2.1.

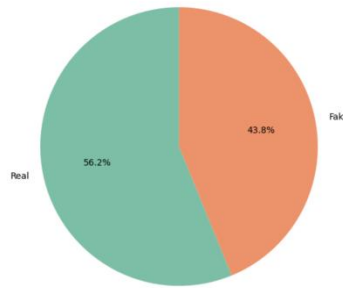


Figure 2.1 Overall Distribution of Real vs Fake Posts

- **Classification Tasks:** In terms of problem formalization, Fakeddit supports both binary and fine-grained classification settings. In particular, the dataset includes a 2-way classification task (True vs. Fake) and a 6-way classification task that distinguishes categories such as satire, misleading content, and imposter content. However, the present EDA focuses only on the 2-way labels, because the binary setting is more suitable for examining broad distribution patterns in subreddit, score, and upvote ratio.
- **Theory:** Figure 2.2 shows fake content clustering in subreddits like r/pareidolia, reflecting Re-framing (Entman, 1993). Creators weaponize visual "reality" to validate textual "falsity." By isolating real images from their original contexts and placing them within misleading textual narratives, creators weaponize the 'reality' of the visual to validate the 'falsity' of the text.

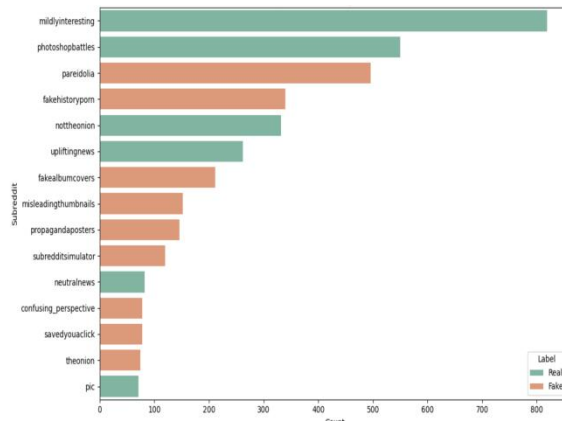


Figure 2.2 Top 15 Subreddits by Post Count

- **Engagement and User Feedback Patterns:**

1. **Visibility (Score):** As shown in Figure 2.3, real posts achieve significantly higher average scores (approx. 764) compared to fake posts (approx. 334). This pattern reflects the episodic virality of genuine news: real posts, drawn from communities covering actual events (political developments, natural disasters, historical records), are capable of generating extreme engagement spikes that inflate the group mean — a phenomenon consistent with the long-tail distribution of content virality. Genuine high-impact events spread broadly across the platform's algorithmic ranking, while fake posts, circulating primarily within pre-converted audiences, tend to plateau at lower absolute engagement. Notably, despite lower scores, fake content still achieves meaningful circulation within its target communities.
2. **Credibility (Upvote Ratio):** Conversely, Figure 2.4 demonstrates that fake posts maintain a higher average upvote ratio (0.886) compared to real posts (0.836). This finding aligns with the Echo Chamber model (Sunstein, 2017): fake and misleading content typically circulates within communities where audience members are predisposed to endorse it, generating uniformly positive reactions. Real content, distributed across a broader and more ideologically diverse audience, accumulates a higher proportion of critical or dissenting votes, depressing the overall upvote ratio. This distinction reveals a paradox: fake content may achieve lower raw visibility but higher within-community endorsement, making it more resistant to correction within its native ecosystem.

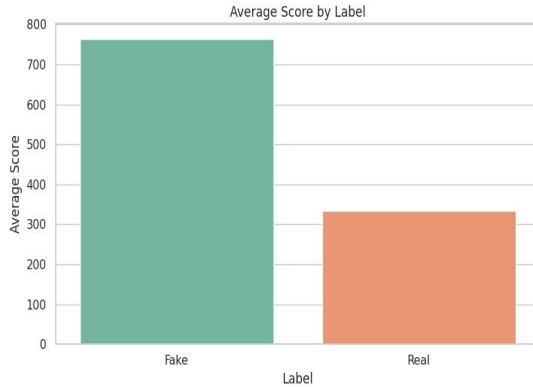


Figure 2.3 Average Score by Label

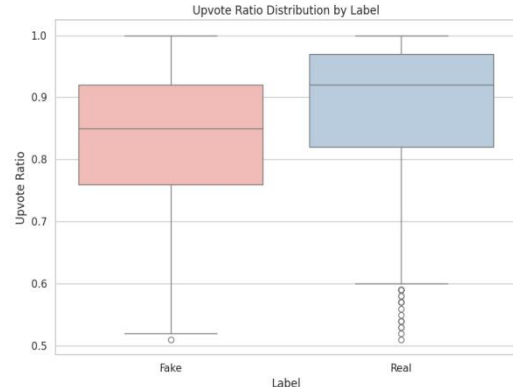


Figure 2.4 Upvote Ratio Distribution by Label

Label

- Conclusion:** Overall, misinformation on Reddit reflects a dual dynamic of reach and reception. Real content achieves greater raw virality (higher scores), spreading beyond its originating community via algorithmic amplification. Fake content, by contrast, cultivates deeper within-community endorsement (higher upvote ratios), embedding itself in echo chambers where shared ideological framing suppresses critical scrutiny. This asymmetry underscores the platform-specific nature of information disorder: virality and credibility signal different, sometimes opposing, dimensions of content performance.

2.2 Model Implementation

- Architecture:** Our model uses a classic "dual-tower" structure. It processes text and images through two separate branches. This class integrates both text and visual features to make a final decision.
 - Text Branch:** We used the bert-base-uncased model to handle the text. For every Reddit post, the title is tokenized into a sequence of 64 tokens. After that the BERTModel outputs a 768-dimensional vector. This vector captures the deep semantic meaning and emotional tone of the post's text.
 - Image Branch:** For the visual part, we chose ResNet18 as the visual feature extractor within the multimodal model. Each input image is encoded into a 512-dimensional feature vector by removing the final classification layer of ResNet18 and treating the remaining layers as a fixed-structure feature extractor. This allows the model to capture mid-level visual representations such as object edges, textures, and structural compositions, without being constrained by ImageNet classification categories.
 - Feature Fusion:** Feature Fusion: This is the core of our model. The 768-dim text representation and the 512-dim image representation are concatenated into a single 1280-dim vector via `torch.cat()`, which is then passed through a two-layer

fully connected classifier. This operation mimics how humans process multimodal information: rather than evaluating text and images in isolation, we integrate both channels to detect cross-modal contradictions that neither modality could expose alone.

- **Cognitive Theory:** Specifically, it aligns with Dual Coding Theory (Paivio, 1971), which posits that the human mind processes verbal and non-verbal information through separate but interconnected channels, making multimodal integration essential for detecting contradictions.

3. Experimental Study & Results

3.1 Quantitative Results

- **Performance Metrics:** The multimodal model demonstrated high performance in binary tasks but faced challenges in fine-grained 6-way classification as detailed in Tables 3.1 and 3.2.. This performance gap shows that although the model performs well in binary detection, it is still challenging to distinguish between specific deception categories.

2-Way Classification (Multimodal)				
[Classification Report]				
	precision	recall	f1-score	support
Real	0.8864	0.9121	0.8991	3457
Fake	0.8576	0.8192	0.8380	2235
accuracy			0.8756	5692
macro avg	0.8720	0.8657	0.8685	5692
weighted avg	0.8751	0.8756	0.8751	5692
Accuracy:	0.8756			
F1 (macro):	0.8685			
F1 (weighted):	0.8751			
AUC (binary):	0.9442			

Table 3.1. 2-Way Classification Report

6-Way Classification (Multimodal)				
[Classification Report]				
	precision	recall	f1-score	support
True	0.8246	0.9065	0.8636	2235
Satire	0.7839	0.4771	0.5932	327
Misleading	0.7519	0.7448	0.7484	1066
Imposter	0.7317	0.2400	0.3614	125
False Connection	0.9451	0.9644	0.9547	1715
Manipulated	0.7407	0.6250	0.6780	224
accuracy			0.8433	5692
macro avg	0.7963	0.6596	0.6999	5692
weighted avg	0.8396	0.8433	0.8356	5692
Accuracy:	0.8433			
F1 (macro):	0.6999			
F1 (weighted):	0.8356			
AUC (macro OvR):	0.9492			

Table 3.2. 6-Way Classification Report

- **Confusion Matrix Analysis:** Figure 3.1 shows that the recognition rate of False Connection is close to perfect (0.96), while the classification error rate of Imposter is serious: 45.6% of the actual imposter content is predicted to be True. There is also an obvious confusion between Satire and True (27.2%).
These classification errors are not arbitrary; they reflect structurally predictable patterns rooted in the communicative logic of each content type.
The Satire–True confusion (27.2%) arises because satirical communities such as r/theonion deliberately adopt the surface conventions of authentic journalism — deploying real photographs, professional-sounding headlines, and credible source aesthetics — while inverting the communicative intent. The model, relying on statistical

associations between visual and textual features and class labels, has no mechanism to detect ironic intent, which is meaningful only within a specific cultural and community context (Entman, 1993). A satirical post and a true news post may share nearly identical multimodal surface features, making the distinction invisible to feature-based classification. This suggests that future systems must incorporate community-context signals (e.g., subreddit identity embeddings) to resolve intent-level ambiguity.

The Imposter category's severe misclassification rate (45.6% predicted as True) reflects a qualitatively different failure mode. Imposter content typically involves legitimate visual material paired with fabricated source attribution — for example, a genuine news photograph falsely credited to a non-existent organization. Since the visual channel carries no anomalous signal, the image branch contributes little discriminative information, and the model defaults toward the majority class. This highlights a systemic limitation of the current architecture: multimodal fusion improves aggregate performance but cannot compensate for cases where one entire modality is deliberately kept inconspicuous. Detecting Imposter content reliably would require external knowledge grounding — such as reverse image search integration or cross-reference with known source databases — which lies beyond the scope of feature-level classification.

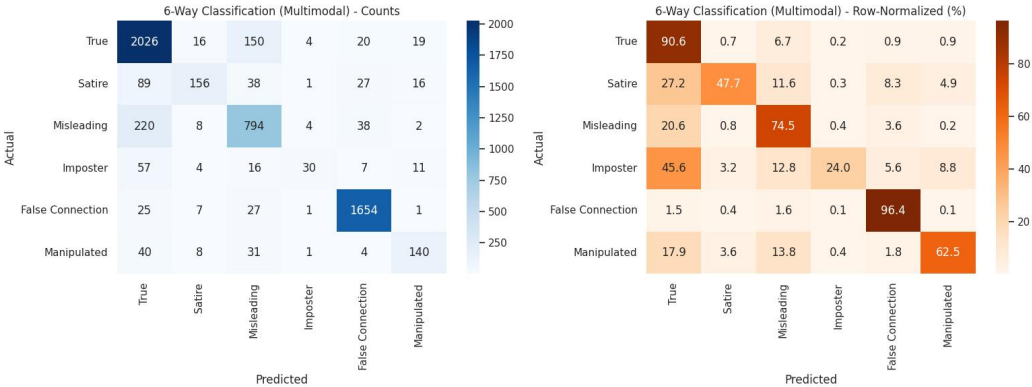


Figure 3.1. 6-Way Evaluation Heat Map

Figure 3.2 shows that False Connection has the highest score (0.96), while Imposter has the lowest score (0.36). The average score of Satire (0.59) and Manipulated (0.68) further confirms that the model has a limited balance between accuracy and recall rate when dealing with fuzzy or tampered content.

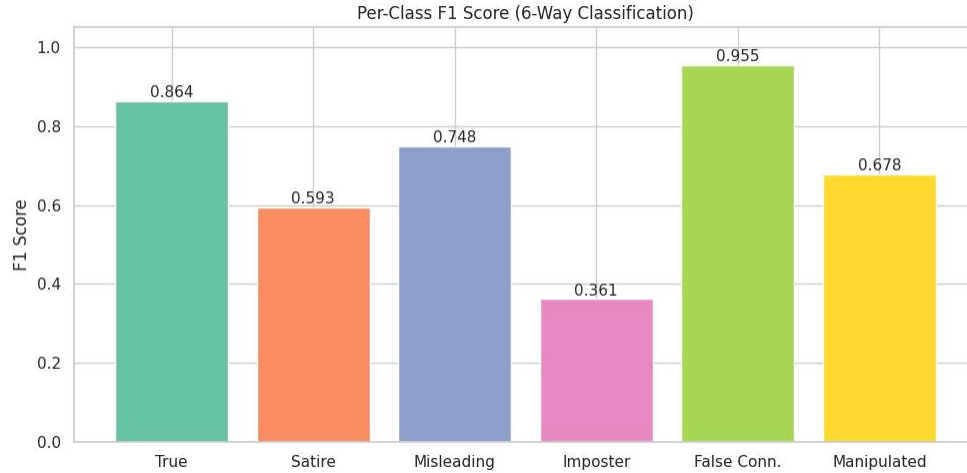


Figure 3.2. Per-Class F1 Bar Chart

3.2 Ablation Study

To quantitatively verify the effectiveness of multimodal fusion, we conduct an ablation study by evaluating three model variants: a text-only model, an image-only model, and the proposed multimodal (text+image) model. The goal is to isolate the contribution of each modality and demonstrate the performance gain achieved through cross-modal integration.

- **Comparative Analysis:**

As shown in Figure 3.3, while the pure text model outperformed the pure image baseline model across all metrics (accuracy: 0.8537 vs. 0.7894; macro F1 score: 0.8456 vs. 0.7841; AUC: 0.9158 vs. 0.8602), the multimodal model achieved the highest scores across all metrics (accuracy: 0.8756; macro F1 score: 0.8685; AUC: 0.9442).

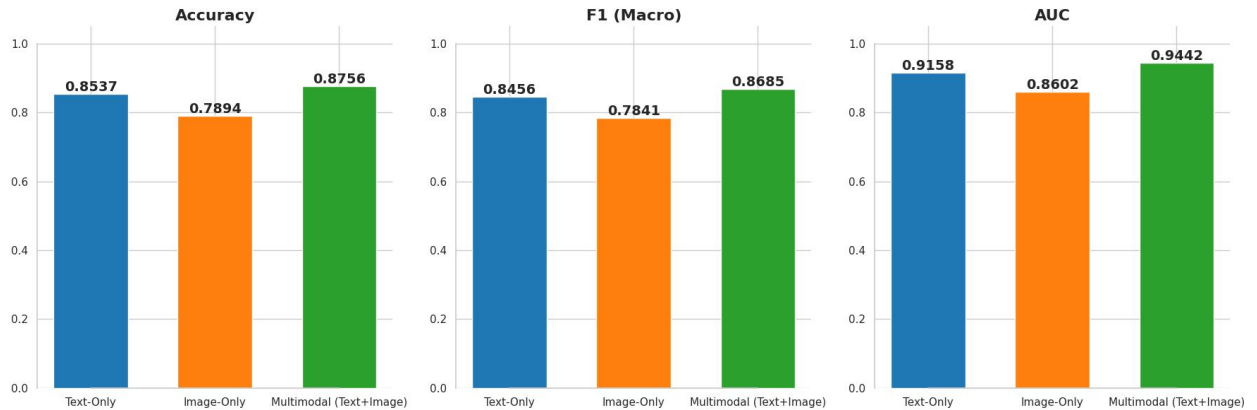


Figure 3.3: Performance Comparison of Different Modal Models

Figure 3.4 confirms that multimodal fusion yields synergistic benefits, significantly outperforming single-modality baselines across all metrics.



Figure 3.4: Performance Gain of Multimodal over Single Modal

- **Finding:** The text-only model achieves reasonable performance, as linguistic patterns such as sensational phrasing, emotional intensifiers, and logical inconsistency provide strong signals for deception. However, image-only performance is notably lower, indicating that visual features alone are insufficient for reliable deception detection. This suggests that while humans often rely on images as Heuristic Cues (Chaiken, 1980) for rapid judgment, visual features alone lack the semantic depth required for high-precision machine detection without textual grounding.

Notably, the gains against the image-only baseline are far larger than those against the text-only baseline across all three metrics, which underscores that visual features alone lack sufficient discriminative power for deception detection, yet become highly valuable when paired with textual information. These results demonstrate that multimodal fusion yields synergistic benefits rather than simple additive gains—text provides the core semantic signal, while images supply complementary visual evidence that resolves ambiguous cases and strengthens the models' confidence in its predictions, as reflected in the significant AUC improvement.

3.3 Interpretability & EDA

- **Community Analysis:** Community Analysis: Our EDA reveals that subreddit membership is one of the strongest predictors of content type and deception mechanism, reflecting the concept of Platform Affordances: each subreddit's community norms, moderation practices, and audience expectations shape not only what content is posted, but how it is encoded and interpreted.

Analysis of the top 15 subreddits by post count reveals sharply polarized fake content ratios. Communities such as r/theonion (fake ratio: 1.0) and r/fakehistoryporn (fake ratio: 1.0) produce exclusively labeled fake content, yet their deception mechanisms differ

fundamentally. *r/theonion* operates as a globally recognized satire outlet — its content is understood by its community as fictional by shared convention, representing a benign but algorithmically problematic form of "fake" content. *r/fakehistoryporn*, by contrast, systematically deploys False Connection (Wardle & Derakhshan, 2017): authentic historical photographs are paired with invented captions, weaponizing visual authenticity to rewrite historical narratives. On the opposite end, *r/neutralnews* (fake ratio: 0.0) enforces strict factual standards, while *r/nottheonion* (fake ratio: 0.0) deliberately collects real events that sound too implausible to be true — inverting the *r/theonion* dynamic. These contrasting community ecologies demonstrate that identical platform infrastructure can simultaneously afford both information integrity and systematic deception, depending entirely on community norms.

The Echo Chamber effect (Sunstein, 2017) amplifies these dynamics. Repeated exposure to community-specific content — whether satirical, conspiratorial, or factual — reinforces shared interpretive frames and raises the threshold for cognitive dissonance. Driven by confirmation bias (Nickerson, 1998), community members develop normalized expectations that disable external critical standards: *r/theonion* readers perceive all content as satirical, making them blind to the rare genuinely deceptive post, while communities with lower epistemic norms may accept fabricated visual narratives as confirmation of pre-existing beliefs. For detection systems, this creates a context-dependency problem that feature-level analysis alone cannot resolve: the same visual and textual surface features may represent benign satire in one subreddit and malicious disinformation in another. This is a key driver of the Satire–True confusion identified in Section 3.1, and motivates the governance recommendation for community-aware content moderation developed in Section 4.3.

- **Visual Attention:**

1. Grad-CAM visualization reveals different attention patterns. For real news, the model focuses on high semantic areas that convey contextual authenticity, such as thick smoke and rolling dust generated by war bombing (e.g., Figure 3.5). This attention to high-arousal visual elements aligns with the Limited Capacity Model (Lang, 2000), which explains how intense visual stimuli can consume human cognitive resources, leading to a neglect of critical text verification.

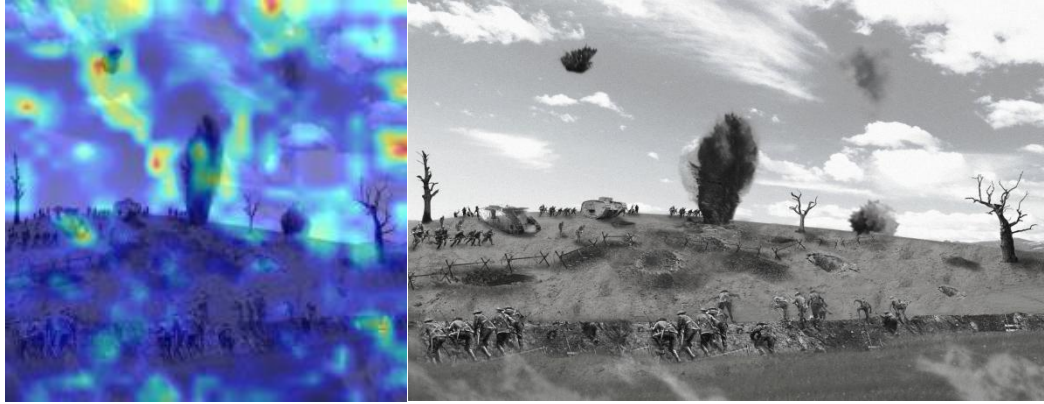


Figure 3.5. Real news gradcam and original picture

2. In contrast, for fake news, attention will be drawn to edge boundaries and local textures, such as the seams of objects processed by Photoshop (e.g., Figure 4). This shows that the model learns to link visual manipulation abnormalities (not semantic content) with deceptive intentions, thus providing key interpretable signals for detection.



Figure 3.6. Fake news gradcam and original picture

4. Application Analysis & Discussion

4.1 Theoretical Mechanisms of Deception

Our analysis of real samples identifies three primary mechanisms of deception:

1. Contextual Re-framing and the "Visual Truth" Trap:

Our analysis identifies Contextual Re-framing as the most sophisticated deception strategy. By pairing authentic images with fabricated narratives, creators weaponize the inherent "realness" of photography to validate falsehoods. Creators weaponize photographic "truth" to validate

fabricated narratives. This Visual Metaphor acts as a heuristic cue, bypassing user skepticism (Chaiken, 1980).

2. Emotional Primacy and Cognitive Load:

The model's focus on high-arousal visual elements aligns with the Limited Capacity Model (LC4MP). Intense visuals (e.g., suffering or sensationalism) consume significant cognitive resources, leaving insufficient mental energy for logical fact-checking. High-arousal visuals consume cognitive resources, leading to the Engagement Paradox: fake posts achieve higher scores as emotional intensity triggers sharing before logical verification (Lang, 2000; Papacharissi, 2015).

3. Ideological Symbolism and the Third-Person Effect:

In specialized subreddits, visual symbols serve as ideological anchors. While our model can detect technical manipulations, the Third-Person Effect (Davison, 1983) suggests that users often overestimate their own immunity to these visual cues. Users may perceive a manipulated image as "harmless satire" while underestimating its long-term impact on public perception. Consequently, multimodal misinformation functions not just as misleading content, but as a tool for reinforcing Echo Chambers, where shared visual framings normalize specific biases and limit the effectiveness of external corrections.

4.2 Case Study Analysis

To ground the theoretical mechanisms identified in Section 4.1 in empirical evidence, we examine two representative cases drawn from the model's actual test outputs, each illustrating a distinct deception strategy.

Case A: Visual Manipulation — r/photoshopbattles (Sample ID: 2k6htd)

Title: "a large frying pan sculpture half submerged in sand"

Subreddit: r/photoshopbattles | True Label: Fake | Prediction: Fake (99.33%)

This post exemplifies Contextual Re-framing through visual fabrication. A physically implausible scene — a monumental frying pan submerged in beach sand — is presented as though it were a real outdoor sculpture. The Grad-CAM heatmap reveals that the model attended primarily to unnatural object boundaries where the digitally inserted element meets the natural background, rather than to any semantically meaningful content. This is consistent with Framing Theory (Entman, 1993): the image imposes an alternative interpretive frame upon reality by embedding a fabricated object in a photorealistic scene. Human viewers employing fast, heuristic processing (Chaiken, 1980) may accept the visual as genuine before engaging in slower logical evaluation — the "Visual Truth" Trap in practice.

Case B: Textual Misdirection — r/mildlyinteresting (Sample ID: cjtvtf)

Title: "my sister and I found a four-leaf and a five-leaf clover"

Subreddit: r/mildlyinteresting | True Label: Fake | Prediction: Fake (99.99%)

This case demonstrates a subtler but equally dangerous form of deception: the image itself is visually unremarkable and contains no detectable manipulation. The deception lies entirely in the textual claim — an extraordinary assertion made in a casual, first-person register that discourages skeptical scrutiny. The model's near-perfect confidence suggests that the textual branch was the primary discriminating signal, detecting linguistic patterns (personal anecdote framing, extraordinary claim, community-appropriate register) associated with misleading content. This reflects the Affective News dynamic (Papacharissi, 2015): content framed as a personal discovery triggers emotional identification and social sharing before triggering critical verification. The Limited Capacity Model (Lang, 2000) further explains why: when cognitive resources are consumed by emotional resonance, critical evaluation is suppressed.

Taken together, these cases demonstrate that multimodal misinformation spans a spectrum — from visually obvious manipulation requiring cross-modal consistency checks (Case A), to semantically subtle misdirection where textual framing alone drives deception (Case B). This spectrum directly motivates the design of our multimodal system: no single-modality model can reliably address both ends of the deception continuum.

4.3 Platform Governance Suggestions

1. The platform should strengthen the consistency detection of text and images, and focus on monitoring high-risk communities such as photoshopbattles and mildlyinteresting.
 2. Establish a graded annotation system: real / satire / misleading / tampered, to avoid misjudging artistic creation as rumors.
 3. Enhance users' media literacy and educate the public to recognize common misleading patterns such as "true picture, false text".
 4. Algorithm governance must adopt a multimodal model and cannot rely solely on the single dimension of text or images.
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5. Conclusion & Implications

5.1 Summary & Limitation

This study confirms that multimodal fusion significantly improves misinformation detection compared to single-modality approaches. However, several limitations remain. First, the model struggles with fine-grained classification, particularly in distinguishing "Imposter" content from "True" posts, suggesting that subtle semantic nuances require more advanced feature extraction. Second, as our analysis is restricted to Reddit-specific data (Fakeddit), the observed engagement patterns may not fully generalize to platforms with different algorithmic logic, such as TikTok or

X. Lastly, the study does not account for the latest Generative AI-driven deceptions (e.g., Deepfakes), which present evolving challenges for detection models.

5.2 Societal Impact

- **Media Literacy:** Media must move beyond technical fixes. Education should empower users to spot "true image, false text" traps. Public awareness must shift from emotional intuition to critical verification. Recognizing visual metaphors is the first line of defense.
 - **Algorithmic Governance:** Platforms must integrate multimodal consistency checks into ranking logic. We recommend a graded labeling system. Distinguishing satire from malicious fraud protects both free expression and factual truth. Proactive moderation should focus on high-risk community clusters. Governance must be systemic, not just reactive.
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Workload Distribution Matrix

Member	Key Responsibilities	Sections
Chen Zitao	Model architecture and data preprocessing.	Methodology / Exp. Study
Zeng Leyao	EDA and social feature analysis (Subreddit/Score).	Introduction / Analysis
Du Zhifei	Model evaluation (6-way metrics and F1 scores).	Experimental Study
Miao Ruoxu	Grad-CAM visualization and visual focus analysis.	Experimental Study
Du yu	Ablation study and single-modal baseline testing.	Methodology / Results
Liu Jiayu	Theory application, case study, and suggestions.	Introduction / Conclusion